

INSPECTION CERTIFICATE EN 10204-3.1

EN 10204:2005-01

Document No.: 465-1509943/001/P

Purchase Order No.: 26682231

Benteler Order No.: 48010538

Dispatch Note No.:

Product: SEAMLESS STEEL TUBES

Inspection No.:

Manufacturer: BSTMC Shreveport USA

Manufacturer's brand:

Stamp of the inspection representative: MI

Steelmaking process: ELECTRIC FURNACE

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DNOW L.P.
7402 N. Eldridge Parkway
HOUSTON TX 77041
USA

Terms of delivery:

ASTM-A 53-2020, ASME SA-53, ASME BPVC.II.A-2021, ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, API Specification 5L
Forty-Sixth Edition April 2018 incl. Errata 1, May 2018, PSL 2, ANSI/NACE MR0175-2021/ISO 15156-1:2020, ANSI/NACE MR0175-2021/ISO
15156-2:2020, ANSI/NACE MR0103/ISO 17945:2015

Dimensions-tolerances:

ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, ASTM-A 53-2020, ASME SA-53, ASME BPVC.II.A-2021, API Specification 5L
Forty-Sixth Edition April 2018 incl. Errata 1, May 2018

Steel grade:

GRADE B, GRADE C, GRADE BN, **GRADE X42N**

Delivery condition:

N

Product marking:

S: BENTELER - BENTELER-Sign - API 5L 1072 - API-Sign - Month/Year - A/SA 53 A/SA 106 NACE - 2,375" OD x
0,218" WT - B/C/BN/X42N - PSL2 - SMLS - Heat No. - Tube length feet - 2.970 - 15038815 - LONGHORN

AEZ = Etching ink marking FK = color marking FS = paint stenciling FSD = Color jet printer FSI = Color jet printer inside LK = Laser marking PKE = tag marking PS = die stamp TS = Ink jet spray marking

Item	Dimensions	Heat No.	Test pressure	
			PSI	sec
0002	2 NPS - SCH 80 18.000 FT - 22.500 FT	320304	2970	5

Country of melt and pour

Melted & Manufactured in USA
Nucor, Jewett, TX, US

Heat analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0002	320304	0.170	0.275	1.00	0.012	0.004	0.19	0.02	0.12	0.13	0.003	0.001	0.002	0.0001

1. Formula: $CE_{IIW} = C + (Mn/6) + ((Cr + Mo + V)/5) + ((Cu + Ni)/15) \leq 0.43 \%$

2. Formula: $Cr + Cu + Mo + Ni + V \leq 1.00 \%$

3. Formula: $Nb + V \leq 0.06 \%$

4. Formula: $Nb + Ti + V \leq 0.15 \%$

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Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0002	320304	0.396	0.463	0.004	0.006

Product analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0002	320304	0.180	0.290	1.00	0.015	0.005	0.19	0.02	0.12	0.14	0.004	0.002	0.003	0.0002
0002	320304	0.180	0.280	0.98	0.013	0.005	0.18	0.02	0.12	0.14	0.003	0.001	0.003	0.0002

1. Formula: $CE_{IIW} = C + (Mn/6) + ((Cr+Mo+V)/5) + ((Cu+Ni)/15) \leq 0.43 \%$

2. Formula: $Cr+Cu+Mo+Ni+V \leq 1.00 \%$

3. Formula: $Nb+V \leq 0.06 \%$

4. Formula: $Nb+Ti+V \leq 0.15 \%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0002	320304	0.407	0.474	0.006	0.009
0002	320304	0.401	0.463	0.004	0.007

Test results

The tubes have been submitted to a leak tightness test by:	Hydrostatic test: acc. to API 5L; Test pressure/Time-record, the test records/charts shall be available for examination by the inspector	PASSED
The tubes are non destructive tested:	ET - Test standard: acc. to API 5L; ET - Test method: acc. to ASTM-E 309; ET - Test drilled hole: 0.125 IN.; ET - Notch shape/hole: drilled hole	PASSED
Visual inspection:	PASSED	
Bending test:	PASSED	
Dimensions examination:	PASSED	

Results of mechanical testing

Tensile test longitudinal at RT, Strip test specimen

Item	Specimen No.	Heat No.	Specimen dimensions	Yield strength	Tensile strength	Elongation	1. Formula	Rt 0.5/Rm
Requirements			IN	RT 0.5 % PSI 42100-71800	Rm PSI 70000-95000	A2" %		MAX 0.93
0002	000011	320304	0.749 X 0.214	55404	81076	33.00	19.00	0.68
0002	001722	320304	0.750 X 0.220	54244	79191	33.00	19.00	0.69

Elongation formula

1. Formula $AF=625000 \cdot AXC^{0,2} / U^{0,9}$

Notched bar impact test [1 CHARPY_V]

Specimen position: longitudinal (L); Test temperature: -50 °F

Item	Specimen No.	Heat No.	Specimen dimensions			Absorbed energy		Impact strength	Shear fracture			
			Length	Width	Height	single	average		single	average	single	average
			mm	mm	mm	ft-lbf	ft-lbf		%	%		
			55	3,33	10,00		MIN 000					
0002	000011	320304	55	3.33	10.00	28			85			
				3.33	10.00	43			85			
				3.33	10.00	32			90	87		

Protection

outside surface varnished

Item	Heat No.	Production No.
0002	320304	15038815

Remark

NACE Standard: Hardness acc. to ANSI/NACE MR0175/ISO 15156: HRC max 22 as per ASTM A370 table 2&3., Hardness acc. to ANSI/NACE MR0103/ISO 17945: HRC max. 22 as per ASTM A370 table 2&3.; Certificate remarks: material free from mercury contamination, Data converted from SI units, No weld repair has been carried out;

Certificate-Remark: